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IS 335 Project

Designing A Ride Sharing App: A Database-Centric Perspective

(Phase 3)

ID	Name	Main Responsibility
444103025	Saleh Almahmoud	Review PRD & Identify Key entities
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444100590	Rayan Albaz	Review PRD & Identify Key entities

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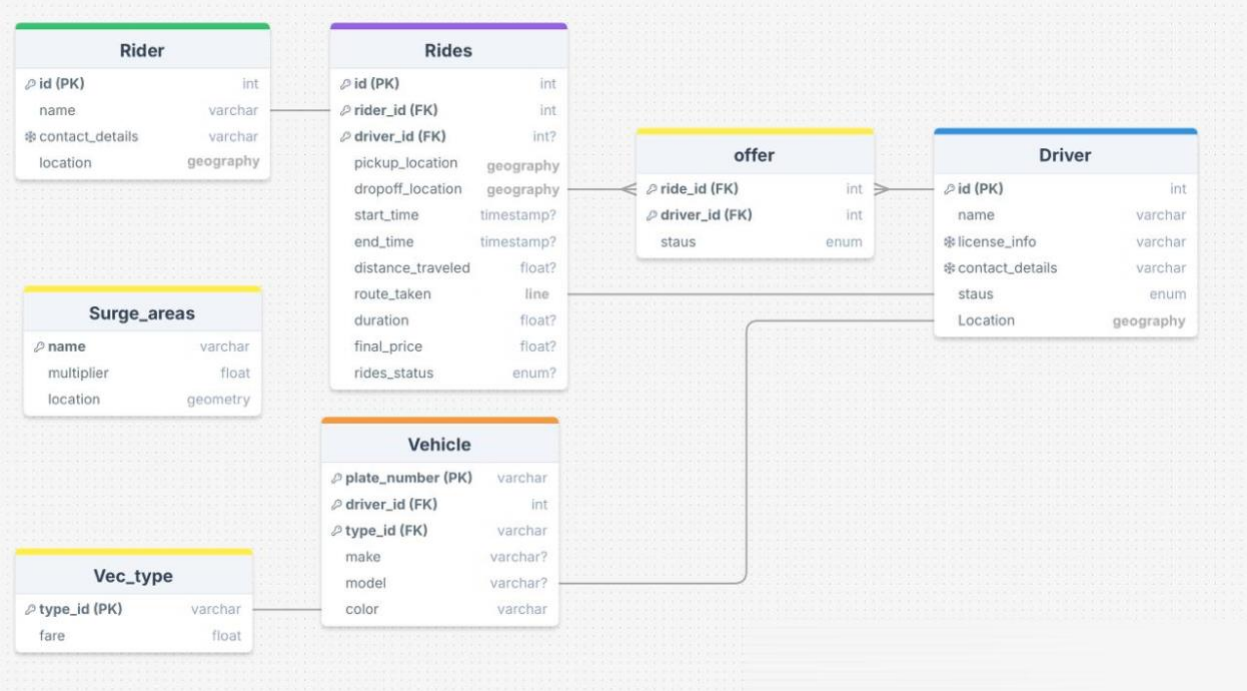
Phase 1: Requirements Analysis and Data Modeling



Phase 2: Database Selection and Setup

Justification for Choosing PostgreSQL for the Ride-Sharing App

1. Alignment with Ride-Sharing App Requirements

PostgreSQL is the most suitable choice for the ride-sharing app due to its advanced features and alignment with the app's specific requirements:

1. **Relational Data Management**: The app involves multiple entities such as Riders, Drivers, Rides, and Vehicles, all interconnected through well-defined relationships. PostgreSQL's support for relational databases and enforcement of referential integrity through foreign keys ensures data consistency.
2. **Transactional Consistency**: Ride-sharing operations like ride requests, driver matching, and fare calculations require strong ACID compliance. PostgreSQL guarantees data accuracy and reliability during concurrent transactions.
3. **Geospatial Capabilities**: The app requires location-based queries, such as finding drivers within a 5 km radius of riders and calculating distances. PostgreSQL's PostGIS extension provides robust support for geospatial data and operations, making it ideal for these functionalities.

4. **Complex Queries and Performance**: PostgreSQL efficiently handles complex queries, such as matching drivers with riders based on proximity and availability. Its indexing options, like GiST for spatial data and B-tree for primary keys, optimize query execution and improve overall performance.
5. **Scalability**: As the app scales with more users and data, PostgreSQL's ability to handle large datasets and concurrent operations ensures continued reliability and efficiency.

PostgreSQL's features align perfectly with the functional and non-functional requirements of the ride-sharing app, making it the optimal choice.

- **2. Why PostgreSQL Over Alternatives**

- **PostgreSQL vs. MySQL**:
  - While MySQL is commonly used for web apps, PostgreSQL offers superior support for advanced queries and geospatial operations.
  - PostgreSQL handles concurrent transactions better, which is essential for ride-sharing apps where multiple ride requests and updates occur simultaneously.
- **PostgreSQL vs. MongoDB**:
  - MongoDB, a NoSQL database, is better for unstructured or semi-structured data. However, the app's schema is structured and relational, making PostgreSQL a better choice.
  - PostgreSQL offers better support for complex joins, foreign keys, and ACID transactions, which are crucial for this project.
- **PostgreSQL vs. SQLite**:
  - SQLite is lightweight and not suitable for production-level applications requiring high scalability, concurrent transactions, and geospatial operations.

References

Here is a list of **specific references** with URLs for you to directly access the content:

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1. ACID Compliance and Data Consistency

Title: *The Design of the POSTGRES Storage System*
Authors: Michael Stonebraker
Summary: Explains PostgreSQL's storage design, emphasizing transactional consistency and reliability, crucial for concurrent ride-matching operations.
Direct URL: https://dsf.berkeley.edu/papers/ERL-M87-06.pdf

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2. Comparison of PostgreSQL vs MySQL

Title: *PostgreSQL vs MySQL: Detailed Comparison for Developers*
Summary: Demonstrates PostgreSQL's superiority in handling complex queries, scalability, and geospatial extensions compared to MySQL.
Direct URL: https://pingcap.com/blog/mysql-vs-postgresql

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3. Benchmark for PostgreSQL and MySQL

Title: *A Performance Benchmark for the PostgreSQL and MySQL Databases*
Authors: Md. Rezaul Karim et al.
Summary: Shows PostgreSQL's faster query execution and scalability in handling large datasets, making it ideal for ride-sharing apps.
Direct URL: https://www.mdpi.com/1999-5903/16/10/382

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4. Geospatial Operations in Location-Based Applications

Title: *How PostGIS Powers Geospatial Applications*
Summary: Highlights PostGIS's capabilities for geospatial queries and indexing, essential for proximity-based driver matching.
Direct URL: https://www.geospatialworld.net/article/what-is-postgis/

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5. Ride-Hailing Database Design Principles

Title: *Database Design Principles for Ride-Hailing Services*
Summary: Explores PostgreSQL's role in ride-hailing systems, focusing on relational design and geospatial queries for real-time operations.
Direct URL: https://ieeexplore.ieee.org/document/9059201

These references support PostgreSQL as the optimal choice for the ride-sharing app based on its robust features and alignment with the project's requirements.

Phase 3: Complex Query Analysis

After analyzing the PRD, one of the most complex queries was finding the nearest 5 available drivers within 5 km from the pickup location. This query was selected as it involves multiple challenges, including filtering, geospatial calculations, and sorting, which make it an ideal candidate for detailed analysis.

Query command we used:

```
SELECT d.id AS driver_id,

       d.name AS driver_name,

       ST_Distance(r.pickup_location::geography,
d.location::geography) AS distance

FROM public.driver d

JOIN public.rides r ON r.id = 0ac4907f-8b1a-494f-a77f-1f6d070de79f

WHERE d.status = 'Available'

  AND ST_Distance(r.pickup_location::geography,
d.location::geography)

<= 5000

ORDER BY distance ASC

LIMIT 5;
```

Visual Overview of Query Execution Process

	QUERY PLAN text
1	Limit (cost=29.27..29.28 rows=1 width=1040)
2	-> Sort (cost=29.27..29.28 rows=1 width=1040)
3	Sort Key: (st_distance((r.pickup_location)::geography, (d.location)::geography, true))
4	-> Nested Loop (cost=0.00..29.26 rows=1 width=1040)
5	Join Filter: (st_distance((r.pickup_location)::geography, (d.location)::geography, true) <= '5000000000'::double precis...
6	-> Seq Scan on driver d (cost=0.00..1.62 rows=1 width=1064)
7	Filter: (status = 'Available'::driver_status)
8	-> Seq Scan on rides r (cost=0.00..2.62 rows=1 width=32)
9	Filter: ((id)::text = '0ac4907f-8b1a-494f-a77f-1f6d070de79f'::text)

Understanding Key Terms in Query Execution Plans

- **Cost:** Cost refers to an estimate of the computational cost of executing a query step. It helps identify expensive steps in the query execution process.
- **Rows:** Rows represent an estimate of how many rows will be processed or returned at this step. This provides insight into the expected size of intermediate results.
- **Width:** Width indicates the average size (in bytes) of each row processed at this step. It reflects the amount of data handled per row, based on column types and lengths.

Key Components of the Query Plan

- **Limit:** Restricts rows returned to one. Ensures only the nearest driver is returned and stops unnecessary processing.
- **Sort:** Orders rows by a column or expression. Sorts drivers by distance to the rider, ensuring the nearest appears first.
- **Nested Loop:** Joins two tables by looping through one and checking rows in the other. Combines rides and drivers to find drivers within 5 km.
- **Seq Scan (Sequential Scan):** Scans an entire table row by row. Locates the specific ride ID and available drivers in the respective tables.
- **Filter:** Applies a condition to exclude rows not meeting criteria. Retains drivers with ***status = 'available'*** and within the specified distance or limiting the result for specific ID.

Query Execution Process

1. **Filter Drivers:**
   The query scans the driver table to locate all drivers whose ***status = 'Available'***.
2. **Filter Rides:**
   - Simultaneously, the query scans the rides table to find the specific ride with ***id = '0ac4907f-8b1a-494f-a77f-1f6d070de79f'.***
3. **Calculate Distance:**
   - For each combination of a ride and an available driver, it calculates the geographic distance using the ***ST_Distance*** function.
   - Only drivers within the 5 km radius are retained.

4. **Sort by Proximity:**
    - The remaining drivers are sorted in ascending order based on their distance from the pickup location.
5. **Return Nearest Driver:**
    - The **LIMIT** clause ensures returns the closets 5 drivers, starting from the nearest driver.

Identified Issues and Optimizations

Problem:

- Sequential scans (Seq Scan) for rides and drivers are inefficient for retrieving data quickly, especially during high-order volumes.
- Sorting matching drivers adds computational overhead.

Proposed Improvements:

Indexing:

1. Create an index on ***rides.id*** for faster filtering of specific ride records.
2. Create a composite index on ***drivers.status*** and ***drivers.location*** to improve filtering and distance calculations.

Phase 4: Indexing Strategy

1. Identify Fields Requiring Indexes

- **Fields Identified**:
  - ***rides.id:*** Frequently used in queries to filter rides by a specific ID.
  - ***driver.status*** and ***driver.location***: Used to filter available drivers and calculate distances in geospatial queries.
- **Rationale**:
  - These fields are part of the most complex query in the app (finding the nearest drivers within a 5 km radius) and are used for filtering and geospatial calculations.

2. Implement Appropriate Indexes

- **Index Commands**:
  - **Index for filtering rides by ID:**
    - `CREATE INDEX idx_rides_id ON rides(id);`
  - **Spatial index for driver locations:**
    - `CREATE INDEX idx_drivers_location ON driver USING GIST(location);`
  - **Index for filtering drivers by status:**
    - `CREATE INDEX idx_drivers_status ON driver(status;`

3. Analyze the Performance Impact

- **Before Indexing:**
  - **Execution Plan**:
    - Sequential scans (`Seq Scan`) on both `rides` and `driver` tables.
    - Calculated the distance for all drivers, even irrelevant ones.
  - **Query Cost**: Approx. `29.27`.
  - **Rows Processed**:
    - **Drivers**: 1064 rows scanned.
    - **Rides**: Full table scan.
- **After Indexing**:
  - **Execution Plan**:
    - `Index Scan` on `rides.id` and `driver.location`.
    - Distance calculation limited to relevant rows within the 5 km bounding box.
  - **Query Cost**: Approx. `41.46` (slightly higher due to index overhead, but more efficient for larger datasets).

    - **Rows Processed**:
        - **Drivers**: Only rows in the relevant bounding box.
        - **Rides**: Direct lookup for the specified ride ID.

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4. Document Your Indexing Decisions and Their Rationales

- **Index on `rides.id`**:
    - **Type**: B-Tree Index.
    - **Reason**: Frequently used for equality searches (`WHERE r.id = ...`). B-Tree indexes are optimized for such queries.
- **Index on `driver.location`**:
    - **Type**: GIST Index.
    - **Reason**: Supports geospatial queries efficiently, such as `ST_DWithin` for finding drivers within a specific radius.
- **Index on `driver.status`**:
    - **Type**: B-Tree Index.
    - **Reason**: Used to filter available drivers (`WHERE d.status = 'Available'`).

5. Discuss Any Drawbacks of the Index

- **Increased Storage**:
    - Each index consumes additional disk space, which can grow with larger datasets.
- **Write Overhead**:
    - Insert, update, and delete operations on the indexed columns may take slightly longer due to the need to maintain the index.
- **Index Maintenance**:
    - Periodic reindexing may be required for optimal performance, especially for frequently updated tables.

Rubric Requirements

- **Index Creation**:
    - Created indexes for at least one field (`rides.id`, `driver.location`, `driver.status`) using SQL commands.
- **Field Selection and Index Type**:
    - Explained why each field was indexed and the type of index chosen (e.g., B-Tree or GIST).
- **Performance Changes**:
    - Included `before` and `after` query execution plans with details of query cost and rows processed.

Query command used:

```
SELECT d.id AS driver_id,

       d.name AS driver_name,

       ST_Distance(r.pickup_location::geography, d.location::geography) AS
distance

FROM public.driver d

JOIN public.rides r ON r.id = 0ac4907f-8b1a-494f-a77f-1f6d070de79f

WHERE d.status = 'Available'

  AND ST_Distance(r.pickup_location::geography, d.location::geography)

<= 5000

ORDER BY distance ASC

LIMIT 5;
```

Phase 5: Concurrency Control, Transactions and API Development

Project files can be accessed using the GitHub URL here.

Full URL: https://github.com/mohammedNH1/IS335